JEE-Main-30-01-2024 (Memory Based)
[EVENING SHIFT]

Physics

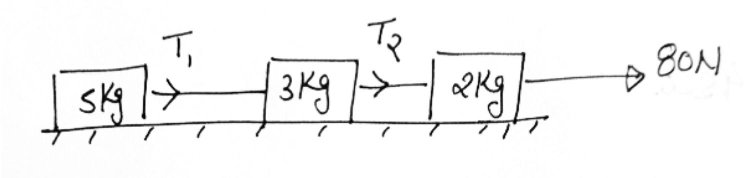
Question: Two polarisers A and B are kept one after the other. Their pass axis makes an angle of 45 with each other. An unpolarized light of intensity I_0 strikes A first and then B. Find the intensity of the light emergent from B.
Options:
(a) $\frac{I_0}{2}$
(b) $\frac{I_0}{4}$
(c) $\frac{I_0}{8}$
(d) $\frac{I_0}{6}$
Answer: (b)

Question: Simple pendulum of length 'l = 4' is taken to height 'R' above earth surface calculate time period at its height {R ⟶ Radius of Earth & Taken $\pi^2 = g$}
Options:
(a) 4 sec
(b) 8 sec
(c) 2 sec
(d) 10 sec
Answer: (b)

Question: If for a given planet $R_P = \frac{1}{3}R_E$ & $M_P = \frac{1}{6}M_E$ then find the escape speed for this planet if the escape speed of earth is 11.2 km/hr
Options:
(a) 7.9 km/hr
(b) 11.2 km/hr
(c) 7.9 m/s
(d) 8.5 m/s
Answer: (a)

Question: A block of mass 1 kg is pulled up on an inclined plane 60º by a force of 10 N. Coefficient of friction is μ = 0.1. Find the magnitude of Work done by the friction by the time block moves up by 10 m.
Options:
(a) 5×10^{-2} J
(b) 5 J
(c) 5000 J
(d) 500 J
Answer: (b)

Question: Find the Tension T_1 & T_2 in the given system



Options:
(a) $T_1 = 40$, $T_2 = 64$
(b) $T_1 = 64$, $T_2 = 40$
(c) $T_1 = 30$, $T_2 = 64$
(d) $T_1 = 40$, $T_2 = 30$
Answer: (a)

Question: Find value of P is the dimensional equation
$M^1 = [C^P G^{-½} h^{1/2}]$
(1) → C → speed of light
(2) → G → Universal gravitational constant
(3) → h → Planck's constant
Options:
(a) 1
(b) ½
(c) -½
(d) -1
Answer: (b)

Question: Charge -q rotating around infinite long wire having charge density ρ at distance r then calculate the time period of that -q Charge.
Options:
(a) $2\pi\sqrt{\frac{mr^2}{2k\rho q}}$
(b) $\pi\sqrt{\frac{mr^2}{2k\rho q}}$
(c) $2\pi\sqrt{\frac{mr^2}{k\rho q}}$
(d) $\pi\sqrt{\frac{mr^2}{k\rho q}}$
Answer: (a)

Question: Match the following

A	$\oint \vec{B} \cdot d\vec{A} = 0$	P	Faraday & lenz's law
B	$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{E_0}$	Q	Gauss law on magnetism
C	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_{(enclosed)}$	R	Ampere's law

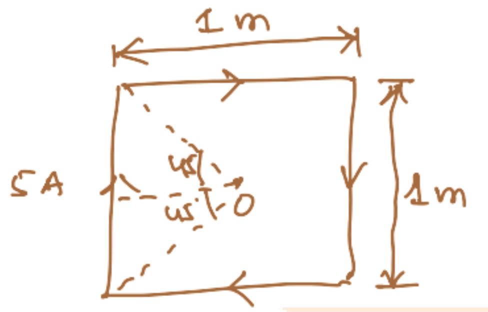
D	$\phi\vec{E}\cdot d\vec{l} = \frac{-d\phi_8}{dt}$	S	Gauss law of Electrostatics

Options:
(a) A → Q, B → S, C → R, D → P
(b) A → S, B → S, C → P, D → R
(c) A → S, B → Q, C → R, D → P
(d) A → Q, B → P, C → S, D → R
Answer: (a)

Question: If 1000 drops of surface energy E_1 are merged to form 1 bigger drop of Surface Energy E_2 then E_1 / E_2 is ______

Question: 2 moles of monatomic gas (γ = 3/2) is mixed with 3 moles of diatomic gas (γ = 5/7).
Find γ of the mixture here $\gamma = C_p/C_V$.
Options:
(a) 5/3
(b) 29/19
(c) 11/7
(d) 39/29
Answer: (b)

Question: Magnetic field at 'O' is $x\sqrt{2} \times 10^{-7}$ Tesla. Find x.



Options:
Answer: (40)

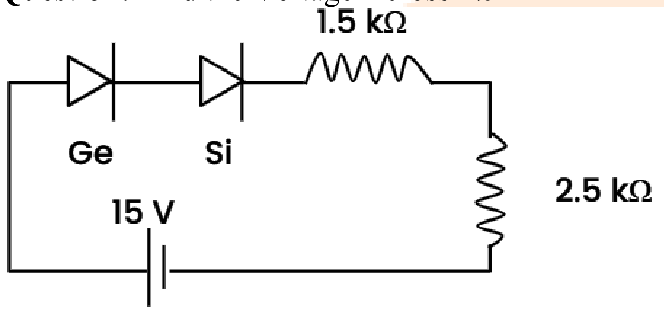
Question: Magnetic moment of electron is proportional to n^P find P
Options:
(a) 3
(b) 2
(c) 4
(d) 1
Answer: (d)

Question: Heat developed in wire is H if wire is cut in 2 equal parts and joined in parallel then new heat dissipated will be?
Options:
(a) H

(b) 2H
(c) 3H
(d) 4H
Answer: (d)

Question: Number of spectral line for He+ for Transition from n = 5 to 1
Options:
(a) 6
(b) 10
(c) 12
(d) 3
Answer: (b)

Question: Find the Voltage Across 2.5 kΩ



Options:
(a) 8.75 V
(b) 7.75 V
(c) 6.75 V
(d) 5.75 V
Answer: (a)

Question: A disc of Moment of inertia 4 kgm^2 is spinning freely at 10 rad/s, a second disc of Moment of Inertia 2 Kgm^2 at angular speed 4 rad.s is put on the first disc and finally they both rotate with same angular speed. What is the change in KE?
Answer: (24)

JEE-Main-30-01-2024 (Memory Based)
[EVENING SHIFT]

Chemistry

Question: Why $KMnO_4$ shows colour ?
Options:
(a) Due to d-d transition
(b) Due to metal to ligand charge transfer
(c) Due to ligand to metal charge transfer
(d) Due to F - center
Answer: (c)
Solution: The transition of a nonbonding 2p oxygen electron to the vacant molecular orbital of a tetrahedral complex causes the lowest energy Ligand to Metal charge transfer.

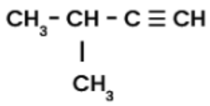
Question: C is added to solution of A and B, find mole fraction of C
Options:
(a) $n_c / (n_A + n_B + n_c)$
(b) $n_c / (n_A - n_B + n_c)$
(c) $n_c / (n_A - n_c + n_B)$
(d) $n_c / (n_A + n_B)$
Answer: (a)
Solution:

$$\text{Mole fraction} = \frac{\text{moles of one component}}{\text{Total moles of all component}}$$

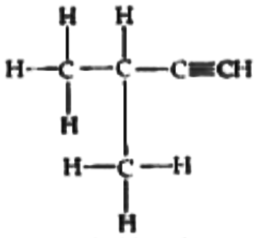
∴ Sum of mole fractions of all components in solution is always one.

$$i.e., x_A + x_B + x_C = \frac{n_A}{n_A + n_B + n_C} + \frac{n_B}{n_A + n_B + n_C} + \frac{n_C}{n_A + n_B + n_C} = 1$$

Question: IUPAC name of compound:

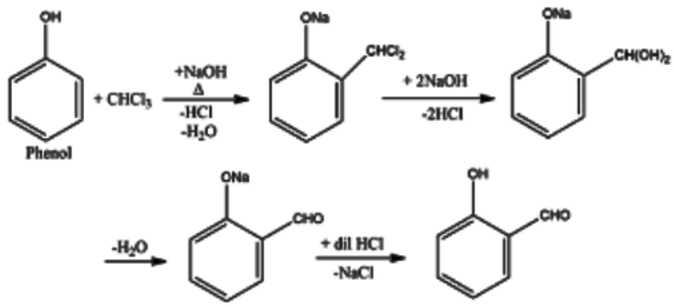


Options:
(a) 2-Methylbutyne
(b) 3-Methylbut-1-yne
(c) 2-Methylbutene
(d) 3-Mehtylbutane
Answer: (b)
Solution:

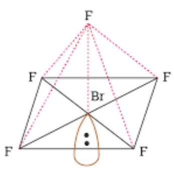


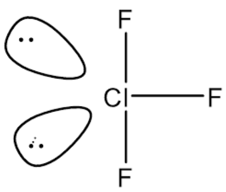
3-Methyl but-1yne

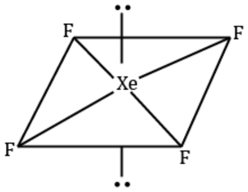
Question: Which reagent on reacting with phenol gives salicylaldehyde ?
Options:
(a) CO_2 , NaOH
(b) $CHCl_3$, NaOH
(c) CCl_4 , NaOH
(d) H_2O, H^+
Answer: (b)
Solution:

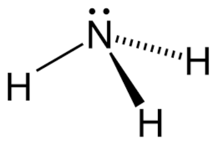


Question: Which of the following has a square pyramidal shape ?
Options:
(a) ClF_3
(b) BrF_5
(c) XeF_4
(d) NH_3
Answer: (b)
Solution:



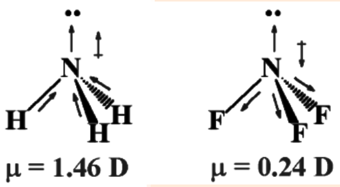




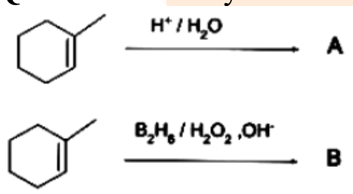


Question: Statement I: NH_3 has lower dipole moment than NF_3

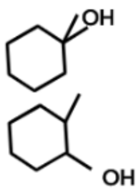
Statement II: In NF_3 flow of electron is in same direction.
Options:
(a) Both statement I and statement II are false
(b) Statement I is true but statement II is false
(c) Statement I is false but statement II is true
(d) Both statement I and statement II are true
Answer: (a)
Solution:



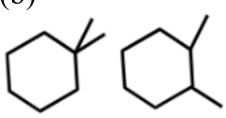
Question: Identify A and B



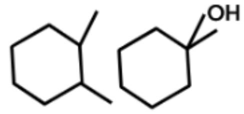
Options:
(a)



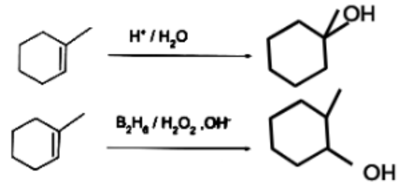
(b)



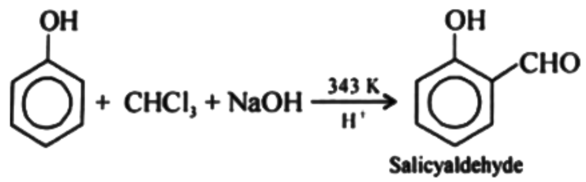
(c)



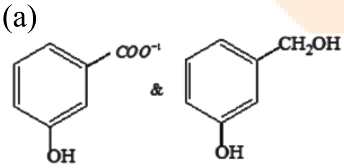
(d) None of the above
Answer: (a)
Solution:



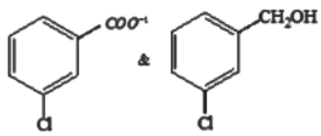
Question: What happens when phenol is treated with chloroform in presence of NaOH at 343 K followed by hydrolysis ?
Options:
(a) Salicylic Acid
(b) Salicylaldehyde
(c) Benzaldehyde
(d) Benzoic acid
Answer: (c)
Solution:



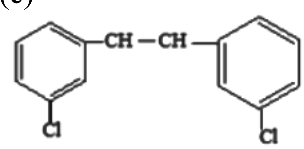
Question: When m - chlorobenzaldehyde is treated with 50% KOH solution, the product(s) obtained is
Options:
(a)



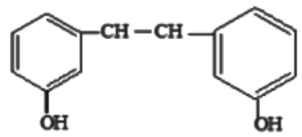
(b)



(c)

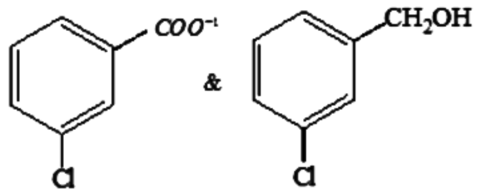


(d)



Answer: (b)

Solution:



Question: Correct IUPAC name the given compound is :

$$CH_3-\underset{\displaystyle NH_2}{\underset{|}{CH}}-CH_2-CN$$

Options:

(a) 3- aminobutanenitrile

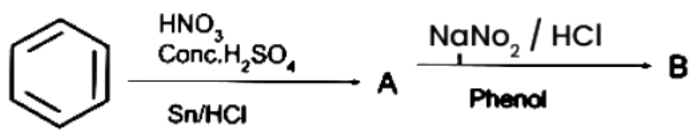
(b) 3- amino cyano butane

(c) 2- aminobutanenitrile

(d) 1- aminobutanenitrile

Answer: (a)

Question: Find B



Options:

(a) p-hydroxy azobenzene

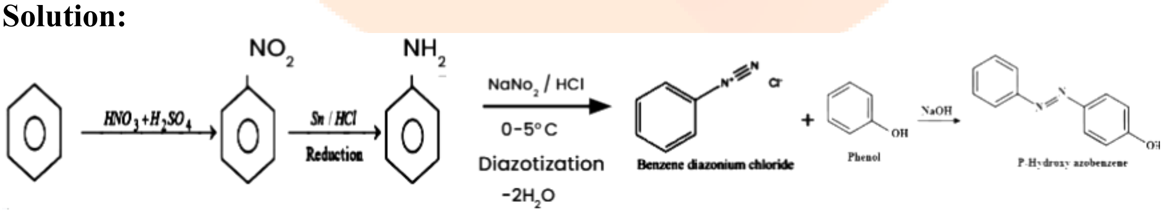
(b) o-hydroxy azobenzene

(c) m-hydroxy azobenzene

(d) Azodye

Answer: (a)

Solution:



Question: In the given reaction A and B respectively are

$CrO_2\ Cl_2 + NaOH \rightarrow A + NaCl + H_2O$

$H_2SO_4 + A + H_2O_2 \rightarrow B$

Options:

(a) Na_2CrO_4, and CrO_5

(b) CrO_5 and Na_2CrO_3

(c) Na_2CrO_4 and CrO_3

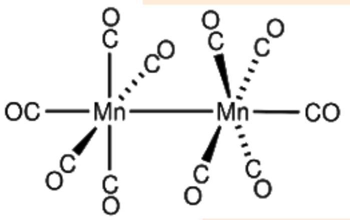
(d) $Na_2 Cr_2 O_7$ and Na_2CrO_4
Answer: (a)
Solution:

$$CrO_2Cl_2 \xrightarrow{NaOH} Na_2CrO_4$$

$$Na_2CrO_4 + H_2O_2 + H_2SO_4 \longrightarrow CrO_5 + Na_2SO_4 + H_2O$$

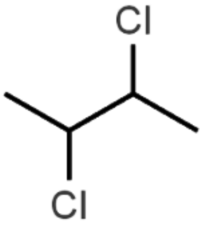
Question: ______ is based on the difference in the solubility of different components of a mixture with a solvent.
Options:
(a) Filtration
(b) Sublimation
(c) Crystallization
(d) Chromatography
Answer: (d)

Question: What is the structure of $Mn_2 (CO)_{10}$?
Options:
(a) Two square pyramidal units joined by bridging CO ligands
(b) Two square pyramidal units joined by Mn-Mn Bond
(c) Two tetrahedral units joined by Mn-Mn Bond
(d) Two square planer units joined by Mn-Mn Bond
Answer: (b)
Solution:

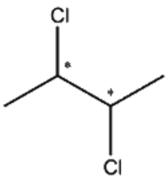


Question: Statement I: H_2 Te is more acidic than $H_2$5
Statement II: H_2 Te has more BDE than $H_2$5
Options:
(a) Statement I and II both are correct
(b) Statement I and II both are incorrect
(c) Statement I is incorrect and Statement II is correct
(d) Statement I is correct and Statement II is incorrect
Answer: (d)
Solution: The bond dissociation enthalpy of H_2 Te is lower than that of H_2 S, and H_2 Te is more acidic

Question: Total optically active compound will show by these compound



Solution:



2, 3 - Dichlorobutane

Number of d and 1 isomers = $2^{n-1} = 2^{2-1} = 2$
Number of meso forms = $2^{n/2-1} = 2^{2/2-1} = 1$
Total number of stereoisomers = 2 + 1 = 3.

Question: Number of spectral lines in the He^+ for transition from n = 5 to n = 1
Solution:
$$\frac{\Delta n(\Delta n+1)}{2} \Rightarrow \frac{4(5)}{2} \Rightarrow 10$$

JEE-Main-30-01-2024 (Memory Based)
[EVENING SHIFT]

Mathematics

Question: Bag A 3 write, 7 Red Ball Bag B → 3 white, 2 red if white ball is found find P of its being from Bag A.

Answer: $\frac{1}{3}$

Solution:

$A \rightarrow 3W,\ TR$

$B \rightarrow 3W,\ 2R$

$$P(W) = \frac{1}{2} \times \frac{3}{10} + \frac{1}{2} \times \frac{3}{5}$$

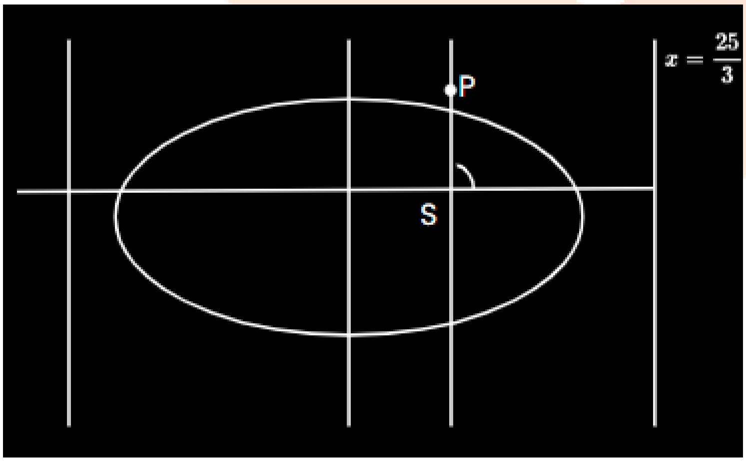
$$P\left(\frac{A}{W}\right) = \frac{\frac{3}{20}}{\frac{3}{20} + \frac{6}{20}} = \frac{1}{3}$$

Question: 5x + 7y = 50

A(A, B), B(O, B) lie in this P divides AB in ratio 7 : 3. $3x - 25 = 0$ is directrix of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with S as its focus on side of directrix if P lies on perpendicular form S to X axis, then find LR of ellipse

Answer: 6.4

Solution:



$5x + 7y = 50$

$A(10,0), B\left(0, \frac{50}{7}\right)$

$P:(3,5)$

$x = \frac{25}{3} = \frac{a}{e}$

$ae = 3;\ a^2 = 245$

$e = \frac{3}{5}$

LR = $\frac{2b^2}{a} = 2a\left(1-e^2\right)$

$= 10\left(\frac{16}{25}\right) = \frac{32}{5} = 6.4$

Question: $A: \frac{x^2}{4} - \frac{x^2}{9} = 1$. If P is point on and area of $\Delta PSS'$; is $2\sqrt{13}$ (S,S' is focus) find square of distance of P from origin.

Answer: $\frac{52}{9}$

Solution:

$a = 2, b = 3$

$e = \sqrt{1+\frac{9}{4}} = \frac{\sqrt{13}}{2}$

$SS' = 2ae$

$= 2\sqrt{13}$

Area $= \frac{1}{2} \times 2\sqrt{13} \times |\beta| = 2\sqrt{3}$

$\Rightarrow |\beta| = 2$

$\frac{\alpha^2}{4} = 1 + \frac{4}{9} = \frac{13}{9}$

$\alpha^2 + \beta^2 = \frac{52}{9} + 4 = \frac{88}{9}$

$\alpha^2 = \frac{52}{9}$

Question: $f(x) = \frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}; g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$. Find $\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)\,dx$

Answer: $\frac{13}{6}$

Solution:

$$f\left(f(x)\right) = \frac{\frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}}{\left(1+\frac{x^4}{1+x^4}\right)^{\frac{1}{4}}}$$

$$= \frac{x}{\left(1+2x^4\right)^{\frac{1}{4}}}$$

$$f\left(f\left(f\left(f\left(x\right)\right)\right)\right) = \frac{x}{\left(1+4x^4\right)^{\frac{1}{4}}}$$

$$\int_0^{\sqrt{2\sqrt{5}}} \frac{x^3}{\left(1+4x^4\right)^{\frac{1}{4}}} dx = \int_1^3 \frac{\frac{t^3}{4} dt}{t} \quad \begin{bmatrix} 1+4x^4 = t^4 \\ 4x^3 dx = t^3 dt \end{bmatrix}$$

$$= \left.\frac{t^3}{2}\right|_1^3 = \frac{13}{6}$$

Question: $f\left(x\right) = ae^{2x} + be^x + cx$

$f\left(0\right) = -1$

$f'\left(\log 2\right) = 21$

$\int_0^{\log_e 4} f\left(x\right) - cx\, dx < \frac{39}{2}$ find $\left|a+b+c\right|$

Answer: 8.00

Solution:

$$f\left(0\right) = a+b = -1 \quad ....(1)$$

$$f'\left(\ln 2\right) = 2a \times 4 + b\left(2\right) + c = 21 \quad ...(2)$$

$$\int_0^{\ln 4} \left(f\left(x\right) - cx\right) dx = \left[\frac{ae^{2x}}{2} + be\right]_0^{\ln 4} = 8a + 4b - \frac{a}{2} - b$$

$$= \frac{15a}{2} + 3b = \frac{39}{2} \quad ...(3)$$

$$\frac{9a}{2} = \frac{45}{2} \Rightarrow a = 5, b = -6, c = -7$$

$$\left|a+b+c\right| = \left|-8\right| = 8$$

Question: Two GP, series (1), $a_1 = a$, $a_3 = b$, series (2) $b_1 = a$, $b_5 = b$. 11th from of series (1) will be which term of series (2)

Answer: 21.00

Solution:

$$a_{11} = a_1 r^{10} = a\left(\frac{b}{a}\right)^5 = \frac{b^5}{a^4}$$

$$b_{n+1} = a \cdot \left(\frac{b}{a}\right)^{\frac{n}{4}} = \frac{b^5}{a^4}$$

$$\Rightarrow n = 20$$

So 21th term.

Question: Given $\left|\vec{b}\right| = 2, \left|\vec{b} \times \vec{a}\right| = 2$. Then $\left|\vec{b} \times \vec{a} - \vec{b}\right|^2$ is

Options:
(a) 0
(b) 8
(c) 1
(d) 10
Answer: (a)
Solution:

$$\left|\vec{b}\right| = 2, \left|\vec{b} \times \vec{a}\right| = 2$$

$$\left|\vec{b} \times \vec{a} - \vec{b}\right|^2 = \left|\vec{b} \times \vec{a}\right|\left|\vec{b}\right|^2$$

$$4 + 4 = 8$$

Question: If $f(x) = \ln\left(\frac{2x}{4x^2 - x - 3}\right) + \cos^{-1}\left(\frac{2x+1}{x+2}\right)$ if domain of $f(x)$ is $[\alpha, \beta)$ then $5\alpha - 4\beta$ is:

Options:
(a) -2
(b) 3
(c) -4
(d) 1
Answer: ()
Solution:

$$\frac{2x}{2x^2 - x - 3} > 0 \Rightarrow \frac{2x}{(2x-3)(x+1)} > 0 \text{ and } -1 \le \frac{2x+1}{x+2} \le 1$$

$$\Rightarrow \frac{2x+1}{x+2} - 1 \le 0 \text{ and } \frac{2x+1}{x+2} + 1 \ge 0$$

$$\frac{x-1}{x+2} \le 0 \text{ and } \frac{3x+3}{x+2} \ge 0$$

$$x \in (-2, 1] \text{ and } x \in (-\infty, -2) \cup [-1, \infty)$$

$$\therefore \text{ Domain } = (-1, 0)$$

$$5\alpha - 4\beta = -5$$

Question: If $f(x)=(x-2)^2(x-3)^3$ and $x\in[1,4]$ of M and m denotes maximum and minimum values respectively, then $M-m$ is
Answer: 12.00
Solution:

$$f'(x)=(x-2)(x-3)^2[2(x-3)+3(x-2)]$$
$$=(x-2)(x-3)^2[5x-12]$$

$$M \Rightarrow f(2)=0 \quad \text{or} \quad f(4)=2^2 1=4$$
$$\Rightarrow M=4$$

$$m \Rightarrow f\left(\frac{12}{5}\right)=0 \quad \text{or} \quad f(1)=-8$$

$$-\frac{108}{325};\ m=-8$$

$$M-m=12$$

Question: Find $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k}$, $|\vec{b}|^2=6$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$. $\vec{a}\cdot\vec{b}=3$ then $(\alpha^2+\beta^2)|\vec{a}\times\vec{b}|^2$ is ____.
Answer: 18.00
Solution:

$$\vec{a}\cdot\vec{b}=|1+\alpha^2+\beta^2|^{\frac{1}{2}}\sqrt{6}\cdot\cos\frac{\pi}{4}=\sqrt{3}|1+\alpha^2+\beta^2|^{\frac{1}{2}}=3$$

$$1+\alpha^2+\beta^2=3$$
$$\Rightarrow \alpha^2+\beta^2=2$$
$$|\vec{a}\times\vec{b}|^2=|\vec{a}|^2|\vec{b}|^2\sin^2\frac{\pi}{4}$$
$$=3\times 6\times\frac{1}{2}=9$$
$$=2\times 9=18$$

Question: If $x\left(x^2+3|x|+5|x-1|+6|x-2|\right)=0,$ then the number of solutions of the given equation is ___.
Answer: 1.00
Solution:

$$x\left[x^2+3|x|+5|x-1|+6|x-21|\right]=0$$

$x=0$ or other expression $=0$

But $x^2, |x|, |x-1|, |x-2|$

So only (1) solution

Question: $3\sin(A+B) = 4\sin(A-B)$. If $\tan A = k\tan B$, then the value of k is ____.

Answer: 7.00

Solution:

$$\frac{\sin(A+B)}{\sin(A-B)} = \frac{4}{3} \to \frac{2\sin A\cos B}{2\cos A\sin B}$$

$$\tan A = 7\tan B$$

$$k = 7$$

Question: If $(y-2)^2 = (x-1)$ and $x-2y+4=0$, then find the area bounded by the curves between the coordinate axis in first quadrant (in sq. units).

Answer: 5.00

Solution:

$$z = zy - 4$$

$$\Rightarrow (y-2)^2 = 2y-5$$

$$y^2 - 6y + 9 = 0$$

$$y = 3$$

$$\text{Area } = \int_0^2 (y-2)^2 + 1dy + \int_2^3 (y-2)^2 + 1 - (2y-4)dy$$

$$= \frac{(y-2)^3}{3} + y\Big|_0^2 + \frac{(y-2)^3}{3}\Big|_2^3 + 5y - y^2\Big|_2^3$$

$$= 2 + \frac{8}{3} + \frac{1}{3} + 5 - 5$$

$$= 5$$

Question: Find the number of common roots of the equation $z^{1901} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$

Answer: 2.00

Solution:

$$z^3 + 2z^2 + 2z + 1 = 0$$

$$(z+1)\left[(z^2 - z + 1) + 2z\right] = 0$$

$$(z+1)(z^2 + z + 1) = 0$$

$$\Rightarrow z = -1,\ w + w^2$$

$z^{1901} + z^{100} + 1 = 0$ has w and w^2 as roots

So 2 common roots

Question: $A=\{1,2,3,4\}$. Find numbers of relations which are symmetric but not reflexive.

Answer: 960.00

Solution:

$(1,1)\quad(1,2)\quad(1,3)\quad(1,4)$

$(2,1)\quad(2,2)\quad(2,3)\quad(2,4)$

$(3,1)\quad(3,2)\quad(3,3)\quad(3,4)$

$(4,1)\quad(4,2)\quad(4,3)\quad(4,4)$

Symmetric = 2^{10}

Reflexive + Symmetric = 2^{6}

Required total = $2^{10}-2^{6}=1024-64=960$